

Analytic Functions

A complex function $w = f(z)$ is said to be analytic at a point z_0 if f is differentiable at z_0 and at every point in some neighborhood of z_0 .

A function f is analytic in a domain D if it is analytic at every point in D . The phrase “analytic on a domain D ” is also used.

Entire Functions

A function that is analytic at every point z in the complex plane is said to be an entire function. In view of differentiation rules (2), (3), (7), and (5), we can conclude that polynomial functions are differentiable at every point z in the complex plane and rational functions are analytic throughout any domain D that contains no points at which the denominator is zero. '

Theorem Polynomial and Rational Functions

- (i) A polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where n is a nonnegative integer, is an entire function.
- (ii) A rational function $f(z) = \frac{p(z)}{q(z)}$, where p and q are polynomial functions, is analytic in any domain D that contains no point z_0 for which $q(z_0) = 0$.

In general, a point z at which a complex function $w = f(z)$ fails to be analytic is called a singular point of f .

Theorem Differentiability Implies Continuity

If f is differentiable at a point z_0 in a domain D , then f is continuous at z_0 .

Proof The limits $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ and $\lim_{z \rightarrow z_0} (z - z_0)$ exist and equal $f'(z_0)$ and 0, respectively.

$$\begin{aligned}\lim_{z \rightarrow z_0} (f(z) - f(z_0)) &= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \cdot (z - z_0) \\ &= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \cdot \lim_{z \rightarrow z_0} (z - z_0) = f'(z_0) \cdot 0 = 0.\end{aligned}$$

From $\lim_{z \rightarrow z_0} (f(z) - f(z_0)) = 0$ we conclude that $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Suppose f and g are functions that are analytic at a point z_0 and $f(z_0) = 0$, $g(z_0) = 0$, but $g'(z_0) \neq 0$. Then

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

EXAMPLE 4 Using L'Hôpital's Rule

Compute $\lim_{z \rightarrow 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i}$.

Solution If we identify $f(z) = z^2 - 4z + 5$ and $g(z) = z^3 - z - 10i$, you should verify that $f(2+i) = 0$ and $g(2+i) = 0$. The given limit has the indeterminate form $0/0$. Now since f and g are polynomial functions, both functions are necessarily analytic at $z_0 = 2+i$. Using

$$f'(z) = 2z - 4, \quad g'(z) = 3z^2 - 1, \quad f'(2+i) = 2i, \quad g'(2+i) = 8 + 12i,$$

we see that

$$\lim_{z \rightarrow 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i} = \frac{f'(2+i)}{g'(2+i)} = \frac{2i}{8 + 12i} = \frac{3}{26} + \frac{1}{13}i.$$

A Necessary Condition for Analyticity

In the next theorem we see that if a function $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point z , then the functions u and v must satisfy a pair of equations that relate their first-order partial derivatives.

Theorem Cauchy-Riemann Equations

Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z = x + iy$. Then at z the first-order partial derivatives of u and v exist and satisfy the **Cauchy-Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \quad (1)$$

Proof The derivative of f at z is given by

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}. \quad (2)$$

By writing $f(z) = u(x, y) + iv(x, y)$ and $\Delta z = \Delta x + i\Delta y$, (2) becomes

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y}. \quad (3)$$

Since the limit (2) is assumed to exist, Δz can approach zero from any convenient direction. In particular, if we choose to let $\Delta z \rightarrow 0$ along a horizontal line, then $\Delta y = 0$ and $\Delta z = \Delta x$. We can then write (3) as

$$\begin{aligned}
f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y) + i[v(x + \Delta x, y) - v(x, y)]}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}.
\end{aligned} \tag{4}$$

The existence of $f'(z)$ implies that each limit in (4) exists. These limits are the definitions of the first-order partial derivatives with respect to x of u and v , respectively. Hence, we have shown two things: both $\partial u/\partial x$ and $\partial v/\partial x$ exist at the point z , and that the derivative of f is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \tag{5}$$

We now let $\Delta z \rightarrow 0$ along a vertical line. With $\Delta x = 0$ and $\Delta z = i\Delta y$, (3) becomes

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}. \quad (6)$$

In this case (6) shows us that $\partial u/\partial y$ and $\partial v/\partial y$ exist at z and that

$$f'(z) = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \quad (7)$$

By equating the real and imaginary parts of (5) and (7) we obtain the pair of equations in (1).

Because Theorem 1 states that the Cauchy-Riemann equations (1) hold at z as a *necessary* consequence of f being differentiable at z , we cannot use the theorem to help us determine where f is differentiable. *But* it is important to realize that Theorem 1 can tell us where a function f does not possess a derivative. If the equations in (1) are *not* satisfied at a point z , then f cannot be differentiable at z .

$f(z) = x + 4iy$ is not differentiable at any point z . If we identify $u = x$ and $v = 4y$, then $\partial u / \partial x = 1$, $\partial v / \partial y = 4$, $\partial u / \partial y = 0$, and $\partial v / \partial x = 0$. In view of

$$\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = 4$$

the two equations in (1) cannot be simultaneously satisfied at any point z . In other words, f is nowhere differentiable.

if a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic throughout a domain D , then the real functions u and v satisfy the Cauchy-Riemann equations (1) at every point in D .

EXAMPLE 1

The polynomial function $f(z) = z^2 + z$ is analytic for all z and can be written as $f(z) = x^2 - y^2 + x + i(2xy + y)$. Thus, $u(x, y) = x^2 - y^2 + x$ and $v(x, y) = 2xy + y$. For any point (x, y) in the complex plane we see that the Cauchy-Riemann equations are satisfied:

$$\frac{\partial u}{\partial x} = 2x + 1 = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}.$$

Criterion for Non-analyticity

If the Cauchy-Riemann equations are not satisfied at every point z in a domain D , then the function $f(z) = u(x, y) + iv(x, y)$ cannot be analytic in D .

EXAMPLE 2 Using the Cauchy-Riemann Equations

Show that the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ is not analytic at any point.

Solution We identify $u(x, y) = 2x^2 + y$ and $v(x, y) = y^2 - x$. From

$$\begin{aligned}\frac{\partial u}{\partial x} &= 4x & \text{and} & & \frac{\partial v}{\partial y} &= 2y \\ \frac{\partial u}{\partial y} &= 1 & \text{and} & & \frac{\partial v}{\partial x} &= -1\end{aligned}\tag{8}$$

we see that $\partial u/\partial y = -\partial v/\partial x$ but that the equality $\partial u/\partial x = \partial v/\partial y$ is satisfied only on the line $y = 2x$. However, for any point z on the line, there is no neighborhood or open disk about z in which f is differentiable at every point. We conclude that f is nowhere analytic.

A Sufficient Condition for Analyticity

By themselves, the Cauchy-Riemann equations do not ensure analyticity of a function $f(z) = u(x, y) + iv(x, y)$ at a point $z = x + iy$. It is possible for the Cauchy-Riemann equations to be satisfied at z and yet $f(z)$ may not be differentiable at z , or $f(z)$ may be differentiable at z but nowhere else. In either case, f is not analytic at z .

However, when we add the condition of continuity to u and v and to the four partial derivatives $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, and $\partial v/\partial y$, it can be shown that the Cauchy-Riemann equations are not only necessary but also sufficient to guarantee analyticity of $f(z) = u(x, y) + iv(x, y)$ at z . The proof is long and complicated and so we state only the result.

Theorem Criterion for Analyticity

Suppose the real functions $u(x, y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in a domain D . If u and v satisfy the Cauchy-Riemann equations (1) at all points of D , then the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in D .

EXAMPLE 3 Using Theorem

For the function $f(z) = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2}$, the real functions $u(x, y) = \frac{x}{x^2 + y^2}$ and $v(x, y) = -\frac{y}{x^2 + y^2}$ are continuous except at the point where $x^2 + y^2 = 0$, that is, at $z = 0$. Moreover, the first four first-order partial derivatives

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{y^2 - x^2}{(x^2 + y^2)^2}, & \frac{\partial u}{\partial y} &= -\frac{2xy}{(x^2 + y^2)^2}, \\ \frac{\partial v}{\partial x} &= \frac{2xy}{(x^2 + y^2)^2}, & \text{and} & \frac{\partial v}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}\end{aligned}$$

are continuous except at $z = 0$. Finally, we see from

$$\frac{\partial u}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{2xy}{(x^2 + y^2)^2} = -\frac{\partial v}{\partial x}$$

that the Cauchy-Riemann equations are satisfied except at $z = 0$. Thus we conclude from Theorem that f is analytic in any domain D that does not contain the point $z = 0$.

In other words, (5) and (7) give us a formula for computing the derivative $f'(z)$:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}. \quad (9)$$

Sufficient Conditions for Differentiability

If the real functions $u(x,y)$ and $v(x, y)$ are continuous and have continuous first-order partial derivatives in some neighborhood of a point z , and if u and v satisfy the Cauchy-Riemann equations (1) at z , then the complex function $f(z) = u(x, y) + iv(x, y)$ is differentiable at z and $f'(z)$ is given by (9).

EXAMPLE 4 A Function Differentiable on a Line

the complex function $f(z) = 2x^2 + y + i(y^2 - x)$ was nowhere analytic, but yet the Cauchy-Riemann equations were satisfied on the line $y = 2x$. But since the functions $u(x, y) = 2x^2 + y$, $\partial u/\partial x = 4x$, $\partial u/\partial y = 1$, $v(x, y) = y^2 - x$, $\partial v/\partial x = -1$ and $\partial v/\partial y = 2y$ are continuous at every point, it follows that f is differentiable on the line $y = 2x$. Moreover, from (9) we see that the derivative of f at points on this line is given by $f'(z) = 4x - i = 2y - i$.

Suppose the function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D .

- (i) If $|f(z)|$ is constant in D , then so is $f(z)$.
- (ii) If $f'(z) = 0$ in D , then $f(z) = c$ in D , where c is a constant.

The Cauchy–Riemann equations also help in deriving general properties of analytic functions.

For instance, show that if $f(z)$ is analytic in a domain D and $|f(z)| = k = \text{const}$ in D , then $f(z) = \text{const}$ in D .

Solution. By assumption, $|f|^2 = |u + iv|^2 = u^2 + v^2 = k^2$. By differentiation,

$$uu_x + vv_x = 0,$$

$$uu_y + vv_y = 0.$$

Now use $v_x = -u_y$ in the first equation and $v_y = u_x$ in the second, to get

$$(a) \quad uu_x - vv_y = 0,$$

$$(b) \quad uu_y + vv_x = 0.$$

To get rid of u_y , multiply (a) by u and (b) by v and add. Similarly, to eliminate u_x , multiply (a) by $-v$ and by u and add. This yields

$$(u^2 + v^2)u_x = 0,$$

$$(u^2 + v^2)u_y = 0.$$

If $k^2 = u^2 + v^2 = 0$, then $u = v = 0$; hence $f = 0$. If $k^2 = u^2 + v^2 \neq 0$, then $u_x = u_y = 0$. Hence, by the Cauchy–Riemann equations, also $v_x = v_y = 0$. Together this implies $u = \text{const}$ and $v = \text{const}$; hence $f = \text{const}$.

Polar Coordinates

As we saw that a complex function can be expressed in terms of polar coordinates. Indeed, the form $f(z) = u(r, \theta) + iv(r, \theta)$ is often more convenient to use. In polar coordinates the Cauchy-Riemann equations become

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}. \quad (10)$$

The polar version of (9) at a point z whose polar coordinates are (r, θ) is then

$$f'(z) = e^{-i\theta} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = \frac{1}{r} e^{-i\theta} \left(\frac{\partial v}{\partial \theta} - i \frac{\partial u}{\partial \theta} \right). \quad (11)$$

Ex: $f(z) = \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r} \cdot e^{-i\theta} = \frac{1}{r}(\cos\theta - i\sin\theta)$

$z \neq 0$.

$u(r, \theta) = \frac{\cos\theta}{r}$ $v(r, \theta) = -\frac{\sin\theta}{r}$ $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$

$r \cdot u_r = -\frac{\cos\theta}{r} = v_\theta$ $u_\theta = -\frac{\sin\theta}{r} = r v_r$

If $z \neq 0$ it has derivative

$f'(z) = e^{-i\theta} \left[-\frac{\cos\theta}{r^2} + i \frac{\sin\theta}{r^2} \right] = e^{-i\theta} \cdot \frac{e^{-i\theta}}{r^2}$

$= -\frac{1}{(re^{i\theta})^2} = -\frac{1}{z^2}$

$r \cdot u_r = v_\theta$

$u_\theta = -r v_r$

$$\forall z \in \mathbb{C}: f(z) = \sqrt[3]{r} e^{i\theta/3} \quad r > 0 \quad \alpha < \theta < \alpha + 2\pi \quad \alpha \in \mathbb{R}$$

$$u(r, \theta) = \sqrt[3]{r} \cos \frac{\theta}{3} \quad v(r, \theta) = \sqrt[3]{r} \sin \frac{\theta}{3}$$

$$\bullet \quad r u_r = \frac{\sqrt[3]{r}}{3} \cos \frac{\theta}{3} = v_\theta \quad u_\theta = -\frac{\sqrt[3]{r}}{3} \sin \frac{\theta}{3} = -r v_r$$

$$f'(z) = e^{-i\theta} \left[\frac{1}{3(\sqrt[3]{r})^2} \cos \frac{\theta}{3} + i \frac{1}{3(\sqrt[3]{r})^2} \sin \frac{\theta}{3} \right]$$

$$f'(z) = \frac{e^{-i\theta}}{3(\sqrt[3]{r})^2} e^{i\theta/3} = \frac{1}{3(\sqrt[3]{r} \cdot e^{i\theta/3})^2} = \frac{1}{3(f(z))^2}$$

The $\partial f / \partial \bar{z} = 0$ form of the Cauchy Riemann equations

Recall that we showed that $f(z) = \bar{z}$ is not analytic anywhere and more generally if $f(z)$ is analytic then $g(z) = f(\bar{z})$ is not complex differentiable at any point for which $f'(\bar{z}) \neq 0$. These observations all suggest that for f to be analytic then it should be possible to express it in a way which does not involve $\bar{z}$. This is indeed the case in the following sense.

From $z = x + iy$ and $\bar{z} = x - iy$ we have

$$x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i}$$

and we can write

$$g(z, \bar{z}) = u(x, y) + iv(x, y) = u\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right) + iv\left(\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2i}\right).$$

For the manipulations we can consider g as independent function of z and $\bar{z}$ and use the chain rule to give

$$\begin{aligned}\frac{\partial g}{\partial \bar{z}} &= \left(\frac{\partial u}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial \bar{z}} \right) + i \left(\frac{\partial v}{\partial x} \frac{\partial x}{\partial \bar{z}} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial \bar{z}} \right) \\ &= \left(\frac{1}{2} \frac{\partial u}{\partial x} - \frac{1}{2i} \frac{\partial u}{\partial y} \right) + i \left(\frac{1}{2} \frac{\partial v}{\partial x} - \frac{1}{2i} \frac{\partial v}{\partial y} \right) \\ &= \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) - \frac{1}{2i} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right).\end{aligned}$$

This shows that

$$\frac{\partial g}{\partial \bar{z}} = 0 \quad \text{if and only if} \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

It is in this sense that if we have a function of z and $\bar{z}$ then it needs to be a function of z alone to be analytic. This does not mean that we can always express an analytic function as a finite combination of standard functions as in all the examples considered so far but as we will show later in the module we can always represent such functions in a power series about a point z_0 which just involves powers of $z - z_0$.

Harmonic Functions

we shall see that when a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic at a point z , then *all* the derivatives of f : $f'(z), f''(z), f'''(z), \dots$ are also analytic at z . As a consequence of this remarkable fact, we can conclude that *all* partial derivatives of the real functions $u(x, y)$ and $v(x, y)$ are continuous at z . From the continuity of the partial derivatives we then know that the second-order mixed partial derivatives are equal. This last fact, coupled with the Cauchy-Riemann equations, will be used in this section to demonstrate that there is a connection between the real and imaginary parts of an analytic function $f(z) = u(x, y) + iv(x, y)$ and the second-order partial differential equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0. \quad (1)$$

This equation, one of the most famous in applied mathematics, is known as **Laplace's equation** in two variables. The sum $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}$ of the two second partial derivatives in (1) is denoted by $\nabla^2 \phi$ and is called the **Laplacian** of ϕ . Laplace's equation is then abbreviated as $\nabla^2 \phi = 0$.

Harmonic Functions A solution $\phi(x, y)$ of Laplace's equation (1) in a domain D of the plane is given a special name.

Definition

A real-valued function ϕ of two real variables x and y that has continuous first and second-order partial derivatives in a domain D and satisfies Laplace's equation is said to be harmonic in D .

Theorem **Harmonic Functions**

Suppose the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D . Then the functions $u(x, y)$ and $v(x, y)$ are harmonic in D .

Proof Assume $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D and that u and v have continuous second-order partial derivatives in D .[†] Since f is analytic, the Cauchy-Riemann equations are satisfied at every point z . Differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to x and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to y give, respectively,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}. \quad (2)$$

With the assumption of continuity, the mixed partials $\partial^2 v/\partial x \partial y$ and $\partial^2 v/\partial y \partial x$ are equal. Hence, by adding the two equations in (2) we see that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{or} \quad \nabla^2 u = 0.$$

This shows that $u(x, y)$ is harmonic.

Now differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to y and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to x , give, in turn, $\partial^2 u/\partial y \partial x = \partial^2 v/\partial y^2$ and $\partial^2 u/\partial x \partial y = -\partial^2 v/\partial^2 x$. Subtracting the last two equations yields $\nabla^2 v = 0$.

EXAMPLE 1 Harmonic Functions

The function $f(z) = z^2 = x^2 - y^2 + 2xyi$ is entire. The functions $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$ are necessarily harmonic in any domain D of the complex plane.

Harmonic Conjugate Functions

We have just shown that if a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its real and imaginary parts u and v are necessarily harmonic in D . Now suppose $u(x, y)$ is a given real function that is known to be harmonic in D . If it is possible to find another real harmonic function $v(x, y)$ so that u and v satisfy the Cauchy-Riemann equations throughout the domain D , then the function $v(x, y)$ is called a **harmonic conjugate** of $u(x, y)$. By combining the functions as $u(x, y) + iv(x, y)$ we obtain a function that is analytic in D .

EXAMPLE 2 Harmonic Conjugate

- (a) Verify that the function $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic in the entire complex plane.
- (b) Find the harmonic conjugate function of u .

Solution

- (a) From the partial derivatives

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial^2 u}{\partial x^2} = 6x, \quad \frac{\partial u}{\partial y} = -6xy - 5, \quad \frac{\partial^2 u}{\partial y^2} = -6x$$

we see that u satisfies Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0.$$

- (b) Since the conjugate harmonic function v must satisfy the Cauchy-Riemann equations $\partial v / \partial y = \partial u / \partial x$ and $\partial v / \partial x = -\partial u / \partial y$, we must have

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 \quad \text{and} \quad \frac{\partial v}{\partial x} = 6xy + 5. \quad (3)$$

Partial integration of the first equation in (3) with respect to the variable y gives $v(x, y) = 3x^2y - y^3 + h(x)$. The partial derivative with respect to x of this last equation is

$$\frac{\partial v}{\partial x} = 6xy + h'(x).$$

When this result is substituted into the second equation in (3) we obtain $h'(x) = 5$, and so $h(x) = 5x + C$, where C is a real constant. Therefore, the harmonic conjugate of u is $v(x, y) = 3x^2y - y^3 + 5x + C$.

In Example 2, by combining u and its harmonic conjugate v as $u(x, y) + iv(x, y)$, the resulting complex function

$$f(z) = x^3 - 3xy^2 - 5y + i(3x^2y - y^3 + 5x + C)$$

is an analytic function throughout the domain D consisting, in this case, of the entire complex plane.

Complex Exponential Function

Definition Complex Exponential Function

The function e^z defined by

$$e^z = e^x \cos y + i e^x \sin y \quad (1)$$

is called the complex exponential function.

$$e^{x+0i} = e^x (\cos 0 + i \sin 0) = e^x (1 + i \cdot 0) = e^x. \quad (2)$$

Theorem Analyticity of e^z

The exponential function e^z is entire and its derivative is given by:

$$\frac{d}{dz} e^z = e^z. \quad (3)$$

EXAMPLE Derivatives of Exponential Functions

Find the derivative of each of the following functions:

(a) $iz^4(z^2 - e^z)$ and (b) $e^{z^2-(1+i)z+3}$.

Solution (a) Using (3) and the product rule

$$\begin{aligned}\frac{d}{dz}[iz^4(z^2 - e^z)] &= iz^4(2z - e^z) + 4iz^3(z^2 - e^z) \\ &= 6iz^5 - iz^4e^z - 4iz^3e^z.\end{aligned}$$

(b) Using (3) and the chain rule

$$\frac{d}{dz}\left[e^{z^2-(1+i)z+3}\right] = e^{z^2-(1+i)z+3} \cdot (2z - 1 - i).$$

Modulus, Argument, and Conjugate The modulus, argument, and conjugate of the exponential function are easily determined from (1). If we express the complex number $w = e^z$ in polar form:

$$w = e^x \cos y + ie^x \sin y = r (\cos \theta + i \sin \theta),$$

then we see that $r = e^x$ and $\theta = y + 2n\pi$, for $n = 0, \pm 1, \pm 2, \dots$. Because r is the modulus and θ is an argument of w , we have:

$$|e^z| = e^x \tag{4}$$

and
$$\arg(e^z) = y + 2n\pi, \quad n = 0, \pm 1, \pm 2, \dots \tag{5}$$

We know from calculus that $e^x > 0$ for all real x , and so it follows from (4) that $|e^z| > 0$. This implies that $e^z \neq 0$ for all complex z . Put another way, the point $w = 0$ is not in the range of the complex function $w = e^z$. Equation (4) does not, however, rule out the possibility that e^z is a negative real number. In fact, you should verify that if, say, $z = \pi i$, then $e^{\pi i}$ is real and $e^{\pi i} < 0$.

A formula for the conjugate of the complex exponential e^z is found using properties of the real cosine and sine functions. Since the real cosine function is even, we have $\cos y = \cos(-y)$ for all y , and since the real sine function is odd, we have $-\sin y = \sin(-y)$ for all y , and so:

$$\overline{e^z} = e^x \cos y - ie^x \sin y = e^x \cos(-y) + ie^x \sin(-y) = e^{x-iy} = e^{\bar{z}}.$$

Therefore, for all complex z , we have shown:

$$\overline{e^z} = e^{\bar{z}}. \tag{6}$$

Theorem Algebraic Properties of e^z

If z_1 and z_2 are complex numbers, then

$$(i) \quad e^0 = 1$$

$$(ii) \quad e^{z_1} e^{z_2} = e^{z_1 + z_2}$$

$$(iii) \quad \frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}$$

$$(iv) \quad (e^{z_1})^n = e^{nz_1}, \quad n = 0, \pm 1, \pm 2, \dots$$

Proof of (i) and (ii) (i) The proof of property (i) follows from the observation that the complex exponential function agrees with the real exponential function for real input. That is, by (2), we have $e^{0+0i} = e^0$, and we know that, for the real exponential function, $e^0 = 1$.

(ii) Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. we have:

$$\begin{aligned} e^{z_1} e^{z_2} &= (e^{x_1} \cos y_1 + i e^{x_1} \sin y_1) (e^{x_2} \cos y_2 + i e^{x_2} \sin y_2) \\ &= e^{x_1+x_2} (\cos y_1 \cos y_2 - \sin y_1 \sin y_2) \\ &\quad + i e^{x_1+x_2} (\sin y_1 \cos y_2 + \cos y_1 \sin y_2). \end{aligned}$$

$$e^{z_1} e^{z_2} = e^{x_1+x_2} \cos(y_1 + y_2) + i e^{x_1+x_2} \sin(y_1 + y_2). \quad (7)$$

$$e^{z_1} e^{z_2} = e^{z_1 + z_2}.$$

Periodicity

The most striking difference between the real and complex exponential functions is the periodicity of e^z . Analogous to real periodic functions, we say that a complex function f is **periodic** with period T if $f(z + T) = f(z)$ for all complex z . The real exponential function is not periodic, but the complex exponential function is because it is defined using the real cosine and sine functions, which are periodic. In particular,

we have $e^{z+2\pi i} = e^z e^{2\pi i} = e^z (\cos 2\pi + i \sin 2\pi)$. Since $\cos 2\pi = 1$ and $\sin 2\pi = 0$, this simplifies to:

$$e^{z+2\pi i} = e^z. \quad (8)$$

The complex exponential function e^z is periodic with a pure imaginary period $2\pi i$.

That is, for the function $f(z) = e^z$, we have $f(z + 2\pi i) = f(z)$ for all z .

Because (8) holds for all values of z we also have $e^{(z+2\pi i)+2\pi i} = e^{z+2\pi i}$. This fact combined with (8) implies that $e^{z+4\pi i} = e^z$. Now by repeating this process we find that $e^{z+2n\pi i} = e^z$ for $n = 0, \pm 1, \pm 2, \dots$. Thus, $-2\pi i, 4\pi i, 6\pi i$, and so on, are also periods of e^z . Furthermore, if the complex exponential function e^z maps the point z onto the point w , then it also maps the points $z \pm 2\pi i, z \pm 4\pi i, z \pm 6\pi i$, and so on, onto the point w . Thus, the complex exponential function is not one-to-one, and all values e^z are assumed in any infinite horizontal strip of width 2π in the z -plane. That is, all values of the function e^z are assumed in the set $-\infty < x < \infty, y_0 < y \leq y_0 + 2\pi$, where y_0 is a real constant. In Figure 1 we divide the complex plane into horizontal strips obtained by setting y_0 equal to any odd multiple of π . If the point z is in the infinite horizontal strip $-\infty < x < \infty, -\pi < y \leq \pi$, shown in color in

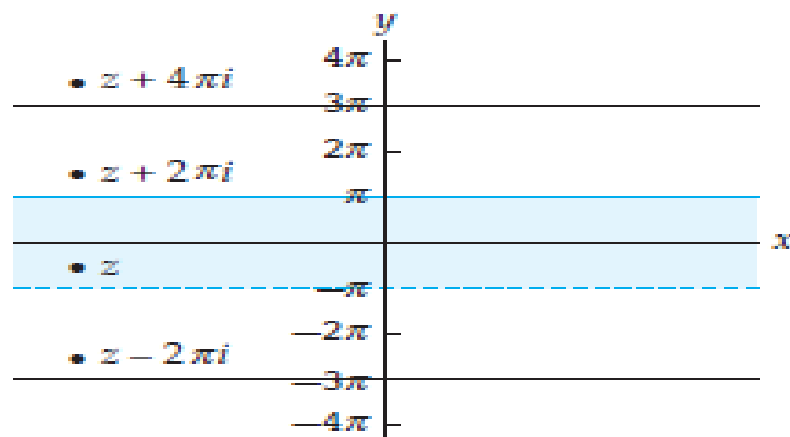


Figure 1, then the values $f(z) = e^z$, $f(z+2\pi i) = e^{z+2\pi i}$, $f(z-2\pi i) = e^{z-2\pi i}$, and so on are the same. The infinite horizontal strip defined by:

$$-\infty < x < \infty, \quad -\pi < y \leq \pi, \quad (9)$$

is called the fundamental region of the complex exponential function.

The Exponential Mapping

Because all values of the complex exponential function e^z are assumed in the fundamental region defined by (9), the image of this region under the mapping $w = e^z$ is the same as the image of the entire complex plane. In order to determine the image of the fundamental region under $w = e^z$, we note that this region consists of the collection of vertical line segments $z(t) = a + it$, $-\pi < t \leq \pi$, where a is any real number.

the image of the line segment $z(t) = a + it$, $-\pi < t \leq \pi$, under the exponential mapping is parametrized by $w(t) = e^{z(t)} = e^{a+it} = e^a e^{it}$, $-\pi < t \leq \pi$,

we see that $w(t)$ defines a circle centered at the origin with radius e^a . Because a can be any real number, the radius e^a of this circle can be any nonzero positive real number. Thus, the image of the fundamental region under the exponential mapping consists of the collection of all circles centered at the origin with nonzero radius. In other words, the image of the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, under $w = e^z$ is the set of all complex w with $w \neq 0$, or, equivalently, the set $|w| > 0$.*

following

The mapping $w = e^z$ of the fundamental region is shown in Figure 1. Each vertical line segment shown in color in Figure 1(a) is mapped onto a circle shown in black in Figure 1(b) by $w = e^z$. As the x -intercept of a vertical line segment increases, the radius of its image increases. Therefore, the leftmost line segment maps onto the innermost circle, the middle line segment maps onto the middle circle, and the rightmost line segment maps onto the outermost circle.

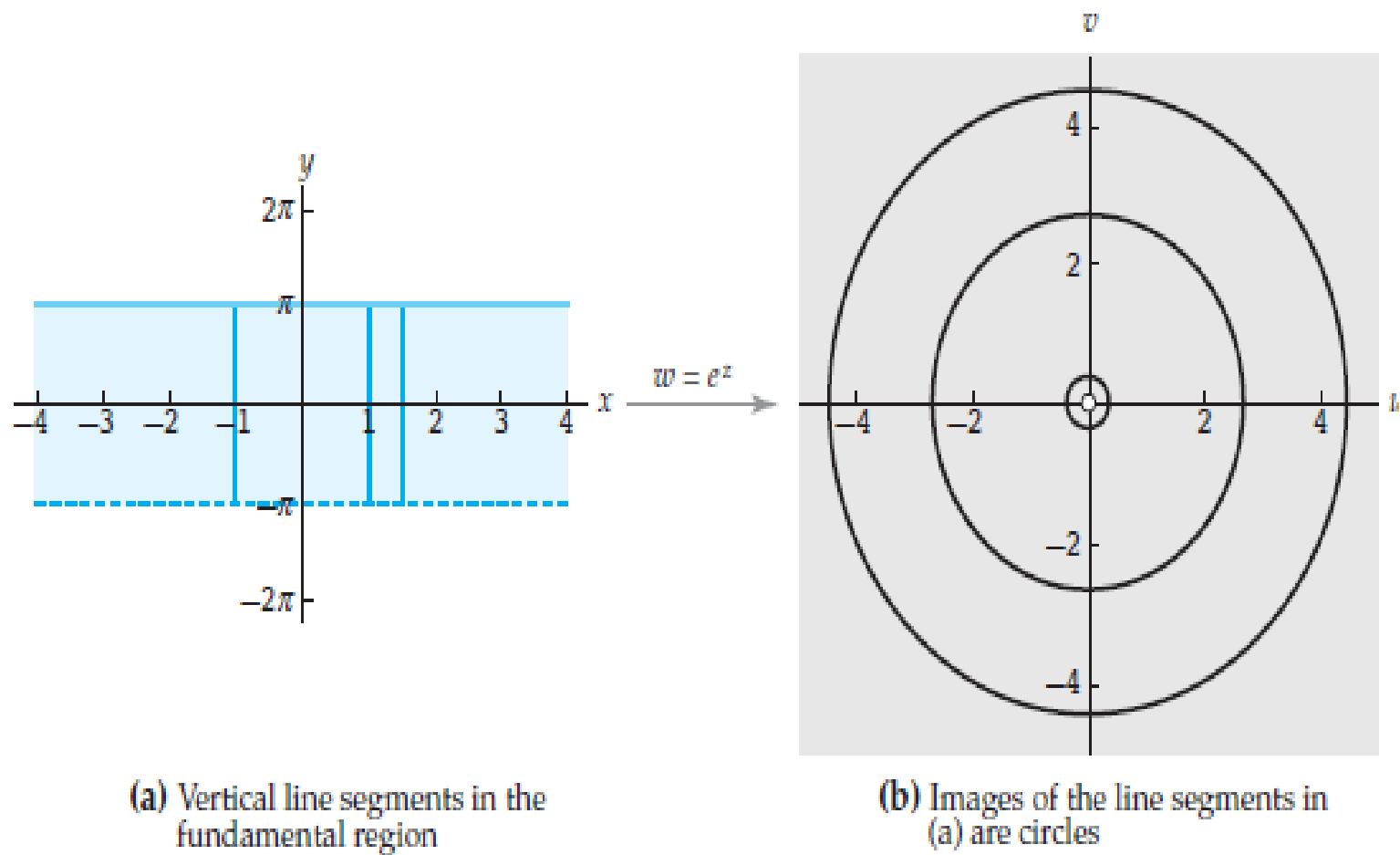
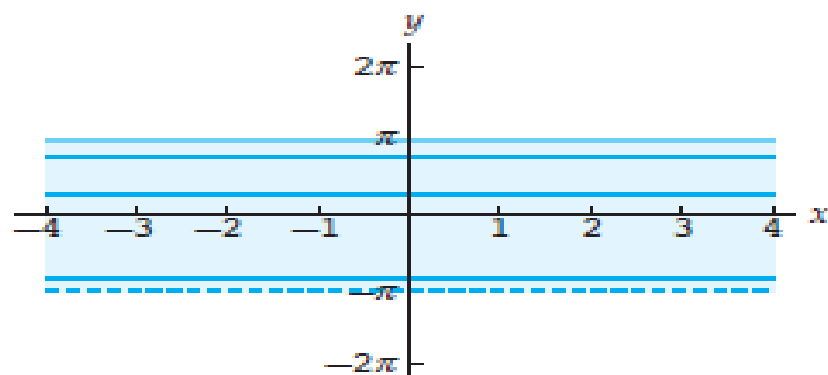


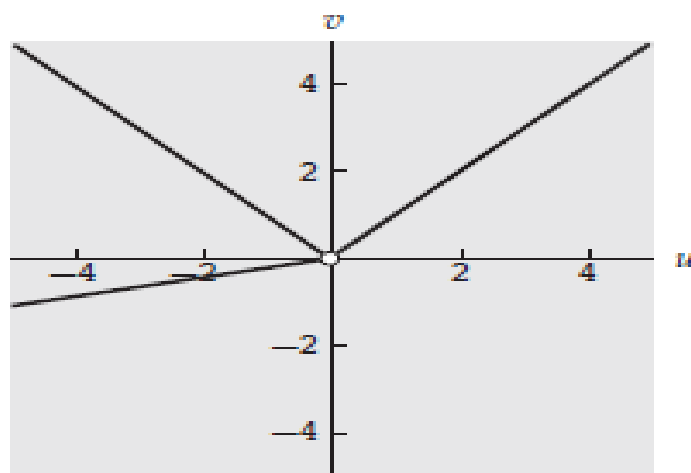
Figure The image of the fundamental region under $w = e^z$

There was nothing particularly special about using vertical line segments to determine the image of the fundamental region under $w = e^z$. The image can also be found in the same manner by using, say, horizontal lines in the fundamental region. In order to see that this is so, consider the horizontal line $y = b$. This line can be parametrized by $z(t) = t + ib$, $-\infty < t < \infty$, and so its image under $w = e^z$ is given by $w(t) = e^{z(t)} = e^{t+ib} = e^t e^{ib}$, $-\infty < t < \infty$. Define a new parameter by $s = e^t$, and observe that $0 < s < \infty$ since $-\infty < t < \infty$. Using the parameter s , the image is given by $W(s) = e^{ib}s$, $0 < s < \infty$, is the set consisting of all points $w \neq 0$ in the ray emanating from the origin and containing the point $e^{ib} = \cos b + i \sin b$. Thus, we have shown that the image of the horizontal line $y = b$ under the mapping $w = e^z$ is the set of all points $w \neq 0$ in the ray emanating from the origin and making an angle of b radians with the positive u -axis. The image can also be described by the equation $\arg(w) = b$. We represent this property of the complex exponential mapping in Figure . Each of the horizontal lines shown in color in Figure (a) is mapped onto a ray shown in black in Figure (b). As the y -intercept of a horizontal line increases, the angle the image ray makes with the positive u -axis increases. Therefore, the bottom-most line maps onto the ray in the third quadrant, the middle line maps onto the ray in the first quadrant, and the top-most line maps onto the ray in the second quadrant.



(a) Horizontal lines in the fundamental region

$w = e^z$



(b) Images of the lines in (a)

Figure The mapping $w = e^z$

Each of the horizontal lines shown in color in Figure (a) is mapped onto a ray shown in black in Figure (b). As the y -intercept of a horizontal line increases, the angle the image ray makes with the positive u -axis increases. Therefore, the bottom-most line maps onto the ray in the third quadrant, the middle line maps onto the ray in the first quadrant, and the top-most line maps onto the ray in the second quadrant.

Exponential Mapping Properties

- (i) $w = e^z$ maps the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, onto the set $|w| > 0$.
- (ii) $w = e^z$ maps the vertical line segment $x = a$, $-\pi < y \leq \pi$, onto the circle $|w| = e^a$.
- (iii) $w = e^z$ maps the horizontal line $y = b$, $-\infty < x < \infty$, onto the ray $\arg(w) = b$.

By using $\omega = f(z) = e^z$, find the mapping of the region $0 \leq x \leq \log_e 2$, $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$ and sketch it.

(a) $0 \leq x \leq \log_e 2$ $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$

$$\omega = e^z = e^{x+iy} = e^x (\cos y + i \sin y) \Rightarrow |\omega| = e^x$$

$$\arg(\omega) = y$$

$$\Rightarrow x=0 \Rightarrow \omega = e^0 = 1$$

$$y = -\frac{\pi}{4} \Rightarrow \arg(\omega) = -\frac{\pi}{4}$$

$$x = \log_e 2 \Rightarrow \omega = e^{\log_e 2} = 2$$

$$y = \frac{\pi}{4} \Rightarrow \arg(\omega) = \frac{\pi}{4}$$



$$\text{If } e^w = z, \text{ then } w = \log_e |z| + i \arg(z). \quad (10)$$

Because there are infinitely many arguments of z , (10) gives infinitely many solutions w to the equation $e^w = z$.

Definition **Complex Logarithm**

The multiple-valued function $\ln z$ defined by:

$$\ln z = \log_e |z| + i \arg(z) \quad (11)$$

is called the **complex logarithm**.

we obtain the following alternative description of the complex logarithm:

$$\ln z = \log_e r + i(\theta + 2n\pi), \quad n = 0, \pm 1, \pm 2, \dots \quad (12)$$

EXAMPLE Solving Exponential Equations

Find all complex solutions to each of the following equations.

(a) $e^w = i$ (b) $e^w = 1 + i$ (c) $e^w = -2$

Solution For each equation $e^w = z$, the set of solutions is given by $w = \ln z$ where $\ln z$ is found using Definition

(a) For $z = i$, we have $|z| = 1$ and $\arg(z) = \pi/2 + 2n\pi$. Thus, from (11) we obtain:

$$w = \ln i = \log_e 1 + i \left(\frac{\pi}{2} + 2n\pi \right).$$

Since $\log_e 1 = 0$, this simplifies to:

$$w = \frac{(4n + 1)\pi}{2}i, \quad n = 0, \pm 1, \pm 2 \dots$$

Therefore, each of the values:

$$w = \dots, -\frac{7\pi}{2}i, -\frac{3\pi}{2}i, \frac{\pi}{2}i, \frac{5\pi}{2}i, \dots$$

satisfies the equation $e^w = i$.

(b) For $z = 1 + i$, we have $|z| = \sqrt{2}$ and $\arg(z) = \pi/4 + 2n\pi$. Thus, from (11) we obtain:

$$w = \ln(1 + i) = \log_e \sqrt{2} + i \left(\frac{\pi}{4} + 2n\pi \right).$$

Because $\log_e \sqrt{2} = \frac{1}{2} \log_e 2$, this can be rewritten as:

$$w = \frac{1}{2} \log_e 2 + \frac{(8n + 1)\pi}{4} i, \quad n = 0, \pm 1, \pm 2, \dots$$

Each value of w is a solution to $e^w = 1 + i$.

(c) Again we use (11). Since $z = -2$, we have $|z| = 2$ and $\arg(z) = \pi + 2n\pi$, and so:

$$w = \ln(-2) = \log_e 2 + i(\pi + 2n\pi).$$

That is,

$$w = \log_e 2 + (2n + 1)\pi i, \quad n = 0, \pm 1, \pm 2, \dots$$

Each value of w satisfies the equation $e^w = -2$.

Theorem Algebraic Properties of $\ln z$

If z_1 and z_2 are nonzero complex numbers and n is an integer, then

$$(i) \quad \ln(z_1 z_2) = \ln z_1 + \ln z_2$$

$$(ii) \quad \ln\left(\frac{z_1}{z_2}\right) = \ln z_1 - \ln z_2$$

$$(iii) \quad \ln z_1^n = n \ln z_1.$$

Proof of (i)

$$\begin{aligned} \ln z_1 + \ln z_2 &= \log_e |z_1| + i \arg(z_1) + \log_e |z_2| + i \arg(z_2) \\ &= \log_e |z_1| + \log_e |z_2| + i (\arg(z_1) + \arg(z_2)). \end{aligned} \quad (13)$$

Because the real logarithm has the property $\log_e a + \log_e b = \log_e(ab)$ for $a > 0$ and $b > 0$, we can write $\log_e |z_1 z_2| = \log_e |z_1| + \log_e |z_2|$. Moreover, we have $\arg(z_1) + \arg(z_2) = \arg(z_1 z_2)$. Therefore, (13) can be rewritten as:

$$\ln z_1 + \ln z_2 = \log_e |z_1 z_2| + i \arg(z_1 z_2) = \ln(z_1 z_2).$$

Principal Value of a Complex Logarithm

It is interesting to note that the *complex* logarithm of a positive real number has infinitely many values. For example, the *complex* logarithm $\ln 5$ is the set of values $1.6094 + 2n\pi i$, where n is any integer, whereas the *real* logarithm $\log_e 5$ has a single value: $\log_e 5 \approx 1.6094$. The unique value of $\ln 5$ corresponding to $n = 0$ is the same as the value of the real logarithm $\log_e 5$. In general, this value of the complex logarithm is called the **principal value of the complex logarithm** since it is found by using the principal argument $\text{Arg}(z)$ in place of the argument $\arg(z)$ in (11). We denote the principal value of the logarithm by the symbol $\text{Ln } z$. Thus, the expression $f(z) = \text{Ln } z$ defines a function, whereas, $F(z) = \ln z$ defines a multiple-valued function.

Definition Principal Value of the Complex Logarithm

The complex function $\text{Ln } z$ defined by:

$$\text{Ln } z = \log_e |z| + i\text{Arg}(z) \quad (14)$$

is called the principal value of the complex logarithm.

$$\text{Ln } z = \log_e r + i\theta, \quad -\pi < \theta \leq \pi. \quad (15)$$

EXAMPLE . Principal Value of the Complex Logarithm

Compute the principal value of the complex logarithm $\text{Ln } z$ for

(a) $z = i$ (b) $z = 1 + i$ (c) $z = -2$

Solution In each part we apply (14) of Definition

(a) For $z = i$, we have $|z| = 1$ and $\text{Arg}(z) = \pi/2$, and so:

$$\text{Ln } i = \log_e 1 + \frac{\pi}{2}i.$$

However, since $\log_e 1 = 0$, this simplifies to:

$$\text{Ln } i = \frac{\pi}{2}i.$$

(b) For $z = 1 + i$, we have $|z| = \sqrt{2}$ and $\text{Arg}(z) = \pi/4$, and so:

$$\text{Ln}(1 + i) = \log_e \sqrt{2} + \frac{\pi}{4}i.$$

Because $\log_e \sqrt{2} = \frac{1}{2} \log_e 2$, this can also be written as:

$$\text{Ln}(1 + i) = \frac{1}{2} \log_e 2 + \frac{\pi}{4}i \approx 0.3466 + 0.7854i.$$

(c) For $z = -2$, we have $|z| = 2$ and $\text{Arg}(z) = \pi$, and so:

$$\text{Ln}(-2) = \log_e 2 + \pi i \approx 0.6931 + 3.1416i.$$

Ln z as an Inverse Function

Because $\text{Ln } z$ is *one* of the values of the complex logarithm $\ln z$, it follows from (10) that:

$$e^{\text{Ln } z} = z \text{ for all } z \neq 0. \quad (16)$$

This suggests that the logarithmic function $\text{Ln } z$ is an inverse function of exponential function e^z . Because the complex exponential function is not

$$\begin{aligned} \text{Ln } e^z &= \log_e |e^z| + i \text{Arg}(e^z) \\ &= \log_e e^x + iy \\ &= x + iy. \end{aligned}$$

Thus, we have shown that:

$$\text{Ln } e^z = z \text{ if } -\infty < x < \infty \quad \text{and} \quad -\pi < y \leq \pi. \quad (17)$$

From (16) and (17), we conclude that $\text{Ln } z$ is the inverse function of e^z defined on the fundamental region. The following summarizes the relationship between these functions.

$\text{Ln } z$ as an Inverse Function of e^z

If the complex exponential function $f(z) = e^z$ is defined on the fundamental region $-\infty < x < \infty, -\pi < y \leq \pi$, then f is one-to-one and the inverse function of f is the principal value of the complex logarithm $f^{-1}(z) = \text{Ln } z$.

Analyticity

$$|z| > 0, \quad -\pi < \arg(z) < \pi, \quad (18)$$

$$f_1(z) = \log_e r + i\theta \quad (19)$$

Theorem Analyticity of the Principal Branch of $\ln z$

The principal branch f_1 of the complex logarithm defined by (19) is an analytic function and its derivative is given by:

$$f_1'(z) = \frac{1}{z}. \quad (20)$$

Logarithmic Mapping The complex logarithmic mapping $w = \text{Ln } z$ can be understood in terms of the exponential mapping $w = e^z$ since these functions are inverses of each other. For example, because $w = e^z$ maps the fundamental region $-\infty < x < \infty$, $-\pi < y \leq \pi$, in the z -plane onto the set $|w| > 0$ in the w -plane, it follows that inverse mapping $w = \text{Ln } z$ maps the set $|z| > 0$ in the z -plane onto the region $-\infty < u < \infty$, $-\pi < v \leq \pi$, in the w -plane. Other properties of the exponential mapping can be similarly restated as properties of the logarithmic mapping. The following summarizes some of these properties.

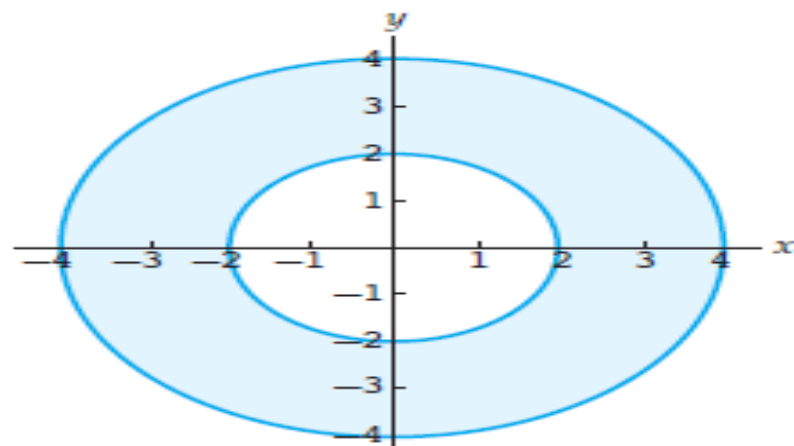
Logarithmic Mapping Properties

- (i) $w = \text{Ln } z$ maps the set $|z| > 0$ onto the region $-\infty < u < \infty$, $-\pi < v \leq \pi$.
- (ii) $w = \text{Ln } z$ maps the circle $|z| = r$ onto the vertical line segment $u = \log_e r$, $-\pi < v \leq \pi$.
- (iii) $w = \text{Ln } z$ maps the ray $\arg(z) = \theta$ onto the horizontal line $v = \theta$, $-\infty < u < \infty$.

EXAMPLE Logarithmic Mapping

Find the image of the annulus $2 \leq |z| \leq 4$ under the logarithmic mapping $w = \text{Ln } z$.

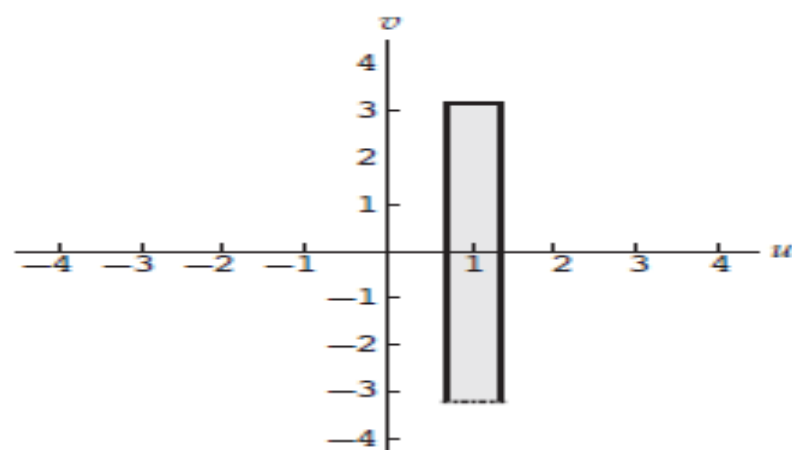
Solution From property (ii) of logarithmic mapping, the boundary circles $|z| = 2$ and $|z| = 4$ of the annulus map onto the vertical line segments $u = \log_e 2$ and $u = \log_e 4$, $-\pi < v \leq \pi$, respectively. In a similar manner, each circle $|z| = r$, $2 \leq r \leq 4$, maps onto a vertical line segment $u = \log_e r$, $-\pi < v \leq \pi$. Since the real logarithmic function is increasing on its domain, it follows that $u = \log_e r$ takes on all values in the interval $\log_e 2 \leq u \leq \log_e 4$ when $2 \leq r \leq 4$. Therefore, the image of the annulus $2 \leq |z| \leq 4$ shown in color in Figure 15.1(a) is the rectangular region $\log_e 2 \leq u \leq \log_e 4$, $-\pi < v \leq \pi$, shown in gray in Figure 15.1(b).



(a) The annulus $2 \leq |z| \leq 4$



$$w = \operatorname{Ln} z$$



(b) The image of the annulus in (a)

Figure The mapping $w = \operatorname{Ln} z$

In each part, determine the image of the set under the given mapping and sketch in the plane.

a) **Sketch** the following region in the complex plane

$$2 < |z| < 4, -\frac{2\pi}{3} < \arg(z) < \frac{2\pi}{3}$$

and

Find the image of the given region under the mapping $\omega = \operatorname{Ln} z$ and sketch it.

b) **Sketch** the following region in the complex plane

$$1 \leq x \leq 2, \frac{\pi}{6} \leq y \leq \frac{\pi}{2}$$

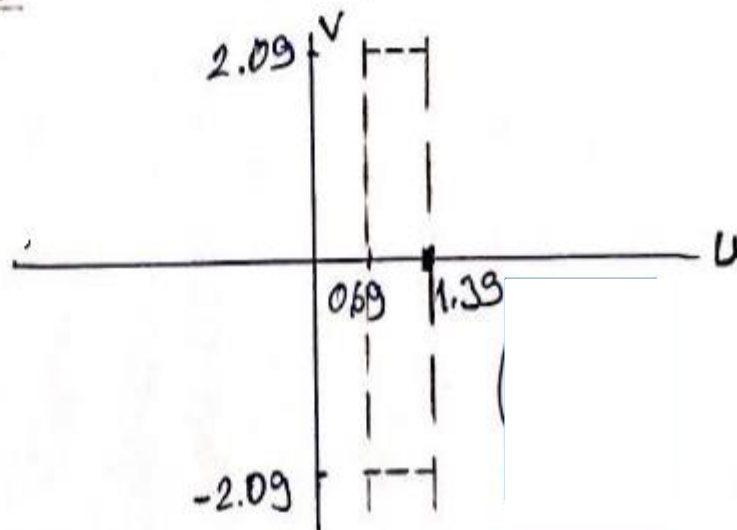
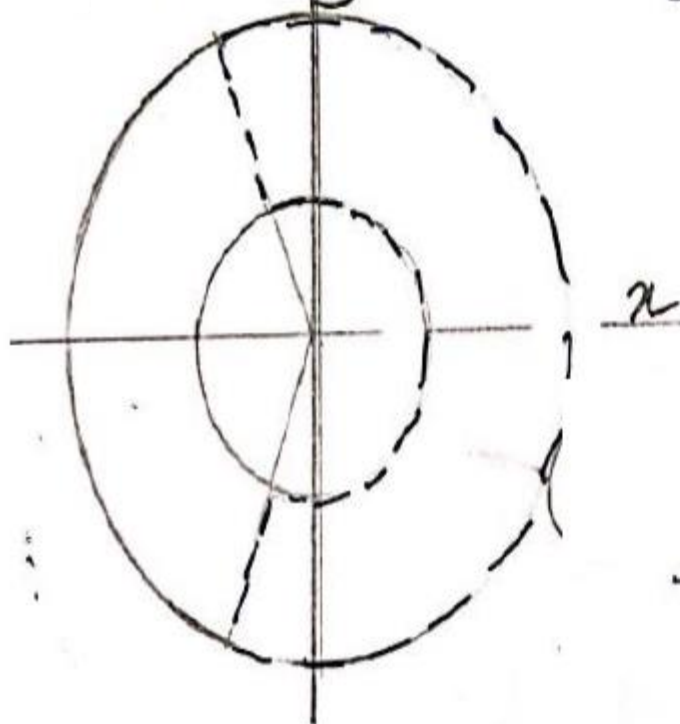
and

Find the image of the given region under the mapping $\omega = e^z$ and sketch it.

a) $2 < |z| < 4$ $-\frac{2\pi}{3} < \arg(z) < \frac{2\pi}{3}$ $\omega = \ln z$

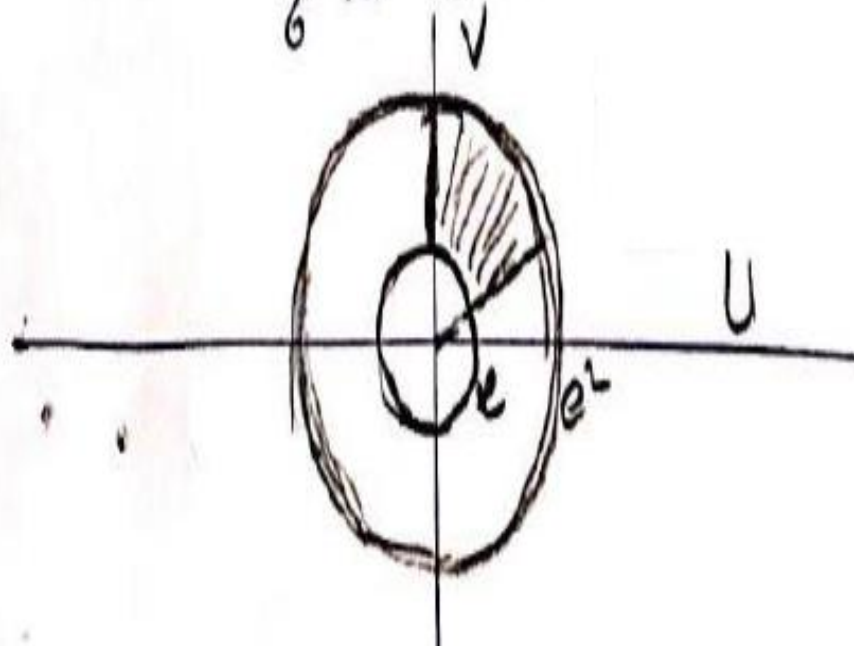
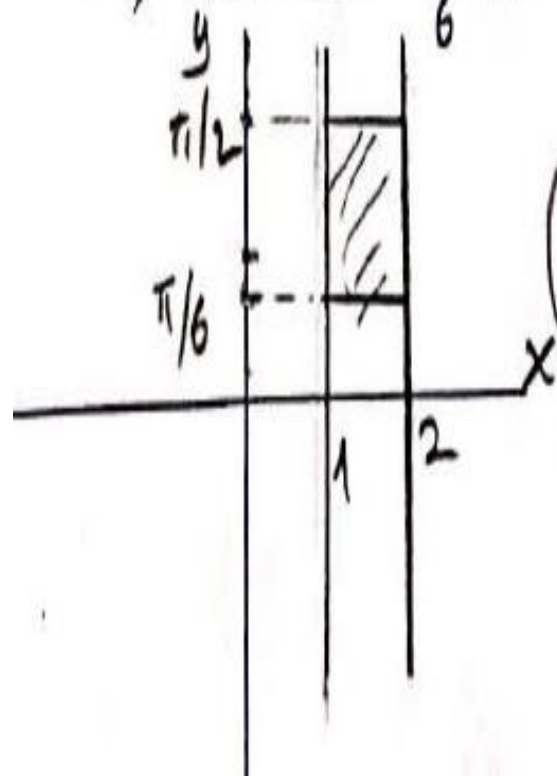
$$\log_e 2 < u < \log_e 4 \Rightarrow 0.69 < u < 1.39$$

$$-\frac{2\pi}{3} < v < \frac{2\pi}{3} \Rightarrow 2.09 < v < -2.09$$



$$b) 1 \leq x \leq 2 \quad \frac{\pi}{6} \leq y \leq \frac{\pi}{2} \quad w = e^t \Rightarrow e^1 \leq |w| \leq e^2$$

$$\frac{\pi}{6} \leq \arg(w) \leq \frac{\pi}{2}$$



Let us remember:

Definition Complex Logarithm

The multiple-valued function $\ln z$ defined by:

$$\ln z = \log_e |z| + i \arg(z)$$

is called the complex logarithm.

Definition 1 Complex Powers

If α is a complex number and $z \neq 0$, then the **complex power** z^α is defined to be:

$$z^\alpha = e^{\alpha \ln z}. \quad (1)$$

$$z^n = e^{n \ln z} = e^{n[\log_e |z| + i \arg(z)]} = e^{n \log_e |z|} e^{n \arg(z)i}. \quad (2)$$

If $\theta = \text{Arg}(z)$, then $\arg(z) = \theta + 2k\pi$ where k is an integer and so

$$e^{n \arg(z)i} = e^{n(\theta + 2k\pi)i} = e^{n\theta i} e^{2nk\pi i}.$$

we have that $e^{2nk\pi i} = \cos(2nk\pi) + i \sin(2nk\pi)$. Because n and k are integers, it follows that $2nk\pi$ is an even multiple of π , and so $\cos(2nk\pi) = 1$ and $\sin(2nk\pi) = 0$. Consequently, $e^{2nk\pi i} = 1$ and (2) can be rewritten as:

$$z^n = e^{n \log_e |z|} e^{n \text{Arg}(z)i}, \quad (3)$$

which is single-valued.

Although the previous discussion shows that (1) can define a single-valued function, you should bear in mind that, in general,

$$z^\alpha = e^{\alpha \ln z} \quad (4)$$

defines a multiple-valued function. We call the multiple-valued function given by (4) a **complex power function**.

EXAMPLE 1 Complex Powers

Find the values of the given complex power: (a) i^{2i} (b) $(1+i)^i$.

Solution In each part, the values of z^α are found using (1).

(a) In part (a) of Example we saw that:

You can find it in the 44. slide

$$\ln i = \frac{(4n+1)\pi}{2}i.$$

Thus, by identifying $z = i$ and $\alpha = 2i$ in (1) we obtain:

$$i^{2i} = e^{2i \ln i} = e^{2i[(4n+1)\pi i/2]} = e^{-(4n+1)\pi}$$

for $n = 0, \pm 1, \pm 2, \dots$. The values of i^{2i} corresponding to, say, $n = -1, 0$, and 1 are 12391.6 , 0.0432 , and 1.507×10^{-7} , respectively.

(b) In part (b) of Example

Slide 45

$$\ln(1+i) = \frac{1}{2} \log_e 2 + \frac{(8n+1)\pi}{4} i$$

for $n = 0, \pm 1, \pm 2, \dots$. Thus, by identifying $z = 1+i$ and $\alpha = i$ in (1) we obtain:

$$(1+i)^i = e^{i \ln(1+i)} = e^{i[(\log_e 2)/2 + (8n+1)\pi i/4]},$$

or

$$(1+i)^i = e^{-(8n+1)\pi/4 + i(\log_e 2)/2},$$

for $n = 0, \pm 1, \pm 2, \dots$.

Let us remember :

Definition **Principal Value of the Complex Logarithm**

The complex function $\text{Ln } z$ defined by:

$$\text{Ln } z = \log_e |z| + i\text{Arg}(z)$$

is called the **principal value of the complex logarithm**.

Complex powers defined by (1) satisfy the following properties that are analogous to properties of real powers:

$$z^{\alpha_1} z^{\alpha_2} = z^{\alpha_1 + \alpha_2}, \quad \frac{z^{\alpha_1}}{z^{\alpha_2}} = z^{\alpha_1 - \alpha_2}, \quad (5)$$

and $(z^\alpha)^n = z^{n\alpha} \quad \text{for } n = 0, \pm 1, \pm 2, \dots$

Definition **Principal Value of a Complex Power**

If α is a complex number and $z \neq 0$, then the function defined by:

$$z^\alpha = e^{\alpha \text{Ln} z} \quad (6)$$

is called the principal value of the complex power z^α .

EXAMPLE 2 Principal Value of a Complex Power

Find the principal value of each complex power: (a) $(-3)^{i/\pi}$ (b) $(2i)^{1-i}$

Solution In each part, we use (6) to find the principal value of z^α .

(a) For $z = -3$, we have $|z| = 3$ and $\text{Arg}(-3) = \pi$, and so $\text{Ln}(-3) = \log_e 3 + i\pi$. Thus, by identifying $z = -3$ and $\alpha = i/\pi$ in (6), we obtain:

$$(-3)^{i/\pi} = e^{(i/\pi)\text{Ln}(-3)} = e^{(i/\pi)(\log_e 3 + i\pi)},$$

or
$$(-3)^{i/\pi} = e^{-1 + i(\log_e 3)/\pi}.$$

$$e^{-1 + i(\log_e 3)/\pi} = e^{-1} \left(\cos \frac{\log_e 3}{\pi} + i \sin \frac{\log_e 3}{\pi} \right),$$

and so,
$$(-3)^{i/\pi} \approx 0.3456 + 0.1260i.$$

(b) For $z = 2i$, we have $|z| = 2$ and $\text{Arg}(z) = \pi/2$, and so $\text{Ln } 2i = \log_e 2 + i\pi/2$
By identifying $z = 2i$ and $\alpha = 1 - i$ in (6) we obtain:

$$(2i)^{1-i} = e^{(1-i)\text{Ln}2i} = e^{(1-i)(\log_e 2 + i\pi/2)},$$

or
$$(2i)^{1-i} = e^{\log_e 2 + \pi/2 - i(\log_e 2 - \pi/2)}.$$

We approximate this value

$$\begin{aligned}(2i)^{1-i} &= e^{\log_e 2 + \pi/2} \left[\cos \left(\log_e 2 - \frac{\pi}{2} \right) - i \sin \left(\log_e 2 - \frac{\pi}{2} \right) \right] \\ &\approx 6.1474 + 7.4008i.\end{aligned}$$

$$f_1(z) = e^{\alpha(\log_e r + i\theta)}, \quad -\pi < \theta < \pi \quad (7)$$

$$f_1'(z) = \frac{d}{dz} e^{\alpha \text{Ln} z} = e^{\alpha \text{Ln} z} \frac{d}{dz} [\alpha \text{Ln} z] = e^{\alpha \text{Ln} z} \frac{\alpha}{z}. \quad (8)$$

$$\frac{d}{dz} z^\alpha = \alpha z^{\alpha-1}. \quad (9)$$

EXAMPLE 3 Derivative of a Power Function

Find the derivative of the principal value z^i at the point $z = 1 + i$.

Solution Because the point $z = 1 + i$ is in the domain $|z| > 0$, $-\pi < \arg(z) < \pi$, it follows from (9) that:

$$\frac{d}{dz} z^i = i z^{i-1},$$

and so,
$$\left. \frac{d}{dz} z^i \right|_{z=1+i} = i z^{i-1} \Big|_{z=1+i} = i (1 + i)^{i-1}.$$

We can use (5) to rewrite this value as:

$$i (1 + i)^{i-1} = i (1 + i)^i (1 + i)^{-1} = i (1 + i)^i \frac{1}{1 + i} = \frac{1 + i}{2} (1 + i)^i.$$

Moreover, from part (b) of Example 1 with $n = 0$, the principal value of $(1 + i)^i$ is:

$$(1 + i)^i = e^{-\pi/4 + i(\log_e 2)/2},$$

and so
$$\left. \frac{d}{dz} z^i \right|_{z=1+i} = \frac{1 + i}{2} e^{-\pi/4 + i(\log_e 2)/2} \approx 0.1370 + 0.2919i.$$

Show where the function $z \mapsto x^3 + i(1 - y)^3$ is:

- (a) analytic;
- (b) differentiable (here $z = x + iy$).

Solution. If the function

$$f(z) = f(x + iy) = x^3 + i(1 - y)^3$$

were analytic then we will have

$$u_x = v_y \quad \Rightarrow \quad 3x^2 = -3(1 - y)^2,$$

which is only satisfied at $x = 0$, $y = 1$; and, on the other hand,

$$u_y = -v_x \quad \Rightarrow \quad 0 = 0,$$

which holds everywhere. Note also that the components of $f(z)$, and all its first-order partials exist everywhere.

Since the Cauchy-Riemann equations only hold at $z = i$, the function $f(z)$ is only differentiable at $z = i$. Hence, in particular, it is not differentiable on any neighbourhood of any point, and therefore is nowhere analytic.

a) Find the values of $(-1 - \sqrt{3}i)^{1/4}$.

b) Sketch the following region on the complex plane

$$|z - 1 + i| < 2, \text{Arg}(z) > \frac{\pi}{2}$$

a) $(-1 - \sqrt{3}i)^{1/4}$ $|-1 - \sqrt{3}i| = 2$ $\arg(-1 - \sqrt{3}i) = \frac{4\pi}{3} + 2n\pi$
 $n \in \mathbb{Z}$

$$\Rightarrow z_n = 2^{1/4} e^{\frac{\frac{4\pi}{3} + 2n\pi}{4}i} \quad n = 0, 1, 2, 3$$

$$\Rightarrow z_0 = 2^{1/4} e^{\frac{\pi}{3}i} = 2^{1/4} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 2^{1/4} \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

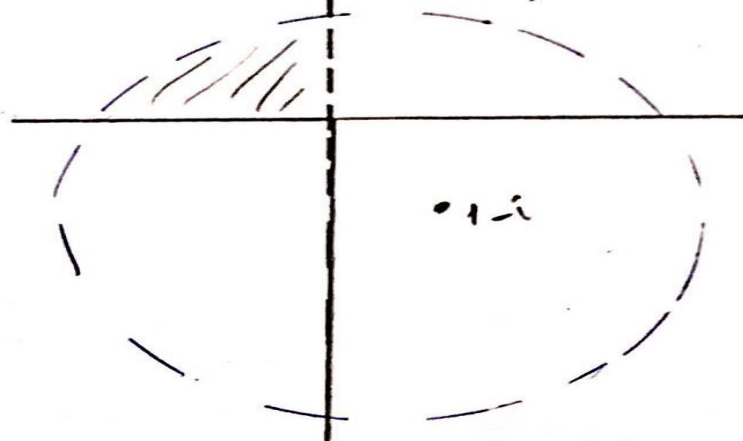
$$z_1 = 2^{1/4} e^{\frac{5\pi}{6}i} = 2^{1/4} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = 2^{1/4} \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$z_2 = 2^{1/4} e^{\frac{4\pi}{3}i} = 2^{1/4} \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) = 2^{1/4} \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$z_3 = 2^{1/4} e^{\frac{11\pi}{6}i} = 2^{1/4} \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right) = 2^{1/4} \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

b) $|z - 1 + i| < 2$ Center: $1 - i$

$\text{Arg } z > \frac{\pi}{2}$ Vertical cap: 2



a) Find all solutions of $e^{2z-1} = i$.

b) . . . Let $z_1, z_2 \in \mathbb{C}$. Show that $\frac{z_1 - z_2}{2\pi i} \in \mathbb{Z}$ is true, if $e^{z_1} = e^{z_2}$.

1) a) $e^{2z-1} = i \quad z = x + iy$

$$\Rightarrow e^{2(x+iy)-1} = e^{2x-1} e^{2iy} = e^{i\frac{\pi}{2}}$$

$$\Rightarrow x = \frac{1}{2} \quad y = \frac{\pi}{4} + n\pi \Rightarrow \boxed{z = \frac{1}{2} + \frac{\pi i}{4} + n\pi i \quad n=0, \pm 1, \pm 2, \dots}$$

$$b) e^{z_1} = e^{z_2} \Rightarrow \frac{e^{z_1}}{e^{z_2}} = 1 \Rightarrow e^{z_1 - z_2} = 1 = e^{2n\pi i}$$

$$\Rightarrow z_1 - z_2 = 2n\pi i \quad n \in \mathbb{Z}$$

$$\frac{z_1 - z_2}{2\pi i} = \frac{2n\pi i}{2\pi i} = n \in \mathbb{Z} \text{ olur.}$$

Let $u(x, y), v(x, y)$ harmonic functions in region D on the plane. And let v is a harmonic conjugate function of u .

a) Show that $u^2 - v^2$ is a harmonic function in D .

And show that $2uv$ is a harmonic conjugate of $u^2 - v^2$.

b) Find the harmonic conjugate function of $u^3 - 3uv^2$.

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow u_{xx} + u_{yy} = 0 \quad \left(\begin{array}{l} u \text{ and } v \\ \text{harmonic fns} \end{array} \right)$$

$$v_{xx} + v_{yy} = 0$$

v is a harmonic conjugate function of u . $\Rightarrow u_x = v_y \quad u_y = -v_x$

a) $\phi = u^2 - v^2 \quad \psi = 2uv$

$$\phi_x = 2uu_x - 2vv_x \quad \phi_{xx} = 2u_x^2 + 2uu_{xx} - 2v_x^2 - 2vv_{xx}$$

$$\phi_y = 2uu_y - 2vv_y \quad \phi_{yy} = 2u_y^2 + 2uu_{yy} - 2v_y^2 - 2vv_{yy}$$

$$\phi_{xx} + \phi_{yy} = 2u(u_{xx} + u_{yy}) - 2v(\underbrace{v_{xx} + v_{yy}}_0) + 2(u_x^2 + u_y^2) - 2(v_x^2 + v_y^2)$$

$$= 2(u_x^2 + u_y^2) - 2(v_x^2 + v_y^2) \quad (u_x = v_y \quad u_y = -v_x)$$

$$= 2(v_y^2 + (-v_x)^2) - 2(v_x^2 + v_y^2) = 0 \Rightarrow \phi = u^2 - v^2$$

is harmonic function

$$\phi = 2uv \quad u_x = v_y \quad u_y = -v_x$$

$$\phi_x = 2u_x v + 2u v_x \quad \phi_y = 2u_y v + 2u v_y$$

$$\phi_x = 2u u_x - 2v v_x \quad \phi_y = 2u u_y - 2v v_y$$

$$\Rightarrow \phi_x = 2v v_y - 2u u_y \Rightarrow \phi_y = -\phi_x$$

$$\phi_y = -2v v_x + 2u u_x \Rightarrow \phi_x = \underline{\phi_y}$$

$\Rightarrow$

$2uv$ is a harmonic conjugate of $u^2 - v^2$

$$b) \quad \psi = u^3 - 3uv^2 \Rightarrow \psi_x = 3u^2u_x - 3u_xv^2 - 6uvu_y$$

$$\psi_y = 3u^2u_y - 3u_yv^2 - 6uvu_x$$

Let $K(x, y)$ is harmonic conjugate function of ψ ,

$$\Rightarrow K_y = \psi_x = 3u^2u_x - 3u_xv^2 - 6uvu_y \quad u_x = v_y \quad u_y = -v_x$$

$$K_y = 3u^2v_y - 3v^2v_y + 6uvu_y \Rightarrow K = 3u^2v - v^3 + g(x)$$

$$-K_x = \psi_y \Rightarrow K_x = -3u^2u_y + 3u_yv^2 + 6uvu_x$$

$$= 3u^2v_x - 3v^2v_x + 6uvu_x$$

$$\Rightarrow 6uvu_xv + 3u^2v_x - 3v^2v_x + \frac{dg}{dx} = 3u^2v_x - 3v^2v_x + 6uvu_x$$

$$\Rightarrow \frac{dg}{dx} = 0 \quad g(x) = A = \text{constant}$$

$$\Rightarrow K(x, y) = 3u^2v - v^3 + A$$

$3u^2v - v^3 + A$ is the harmonic conjugate function of $u^3 - 3uv^2$.

D is a region on the complex plane.

Let $D^* = \{\bar{z} \mid \forall z \in D\}$. If $f(z): D \rightarrow \mathbb{C}$ is a differentiable function, $f^*: D^* \rightarrow \mathbb{C}$, show that $f^*(z) = \overline{f(\bar{z})}$ is differentiable.

$$f(z) = f(x+iy) = u(x,y) + i v(x,y) \quad \text{is differentiable in } D$$

$$u_x = v_y \quad u_y = -v_x \quad \text{Cauchy-Riemann equations are satisfied}$$

$$\begin{aligned} f^*(z) = \overline{f(\bar{z})} &= \overline{f(x-iy)} = \overline{u(x,-y) + i v(x,-y)} \\ &= \tilde{u}(x,y) + i \tilde{v}(x,y) \end{aligned}$$

$$\begin{aligned} \Rightarrow \tilde{u}(x,y) &= u(x,-y) \\ \tilde{v}(x,y) &= -v(x,-y) \end{aligned}$$

$$\Rightarrow \tilde{u}_x = u_x \quad \tilde{u}_y = -u_y \quad \tilde{v}_x = -v_x \quad \tilde{v}_y = v_y$$

$$\begin{aligned} u_x &= v_y \Rightarrow \tilde{u}_x = \tilde{v}_y \\ u_y &= -v_x \Rightarrow -\tilde{u}_y = \tilde{v}_x \end{aligned} \quad \left. \vphantom{\begin{aligned} u_x &= v_y \\ u_y &= -v_x \end{aligned}} \right) \text{Cauchy-Riemann equations are satisfied}$$

$$\Rightarrow f^*(z) \quad \text{is analytic.}$$